California College of the Arts

Systems Design

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Narrative Object Models

Overview

- It's essential to design from two points of view: the structural perspective and the task perspective.
    - Task perspective artifacts include storyboards, journey maps, scenarios, and prototypes.
    - Structural perspective artifacts include a system's overall architecture, system vocabulary, and global design conventions.
- The focus of the readings from this book is Object Modeling and Task Flow Definition.
- **Unified Modeling Language (UML)** is a notational system created in the 1990s to represent system design using a standard set of diagrams.
- There are two main types of UML models: **behavioral** and **structural**.
- **Narrative object model** (mostly) dispenses with the UML notation and instead relies on a plain-English narrative.

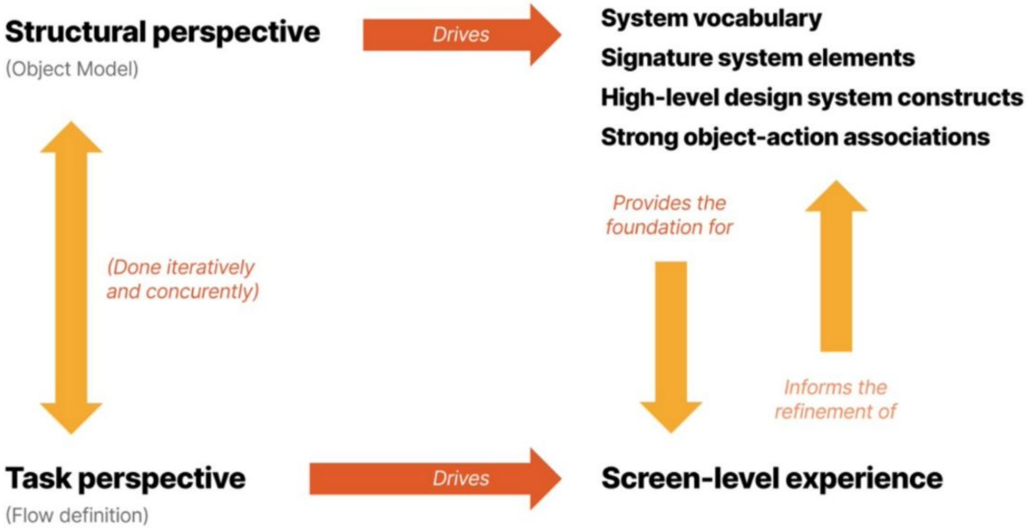


Figure 1.1

Designing from two perspectives (the structural perspective and the task perspective)

Instructions

Here are the 3 main steps:

1) Identifying objects
2) Characterizing the relationships between objects
3) Identifying each object's actions and attributes

We're going to circle back to relationships and focus on the object-action pairs that represents a task in the system.

It’s important to understand the difference between an object, which represents a concept, and an instance of an object. In Figure 2.11, *Customer* is the object, and the instances of that object are the specific database records for Jose E. Hamill, Jamie Averett, and Tiffany Scott.

Figure 2.11

The Customer object and example instances of that object

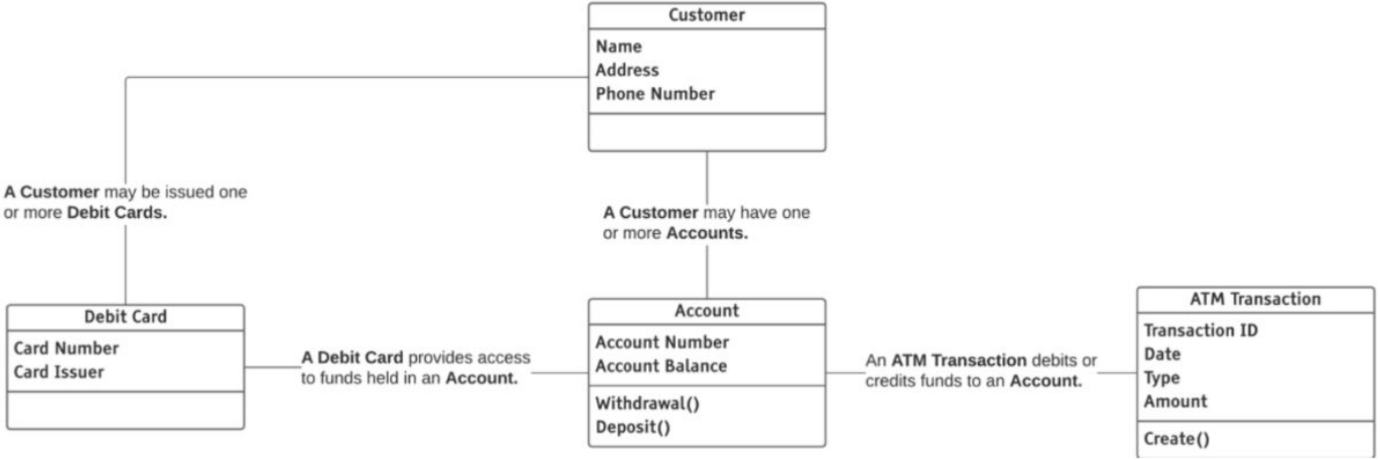
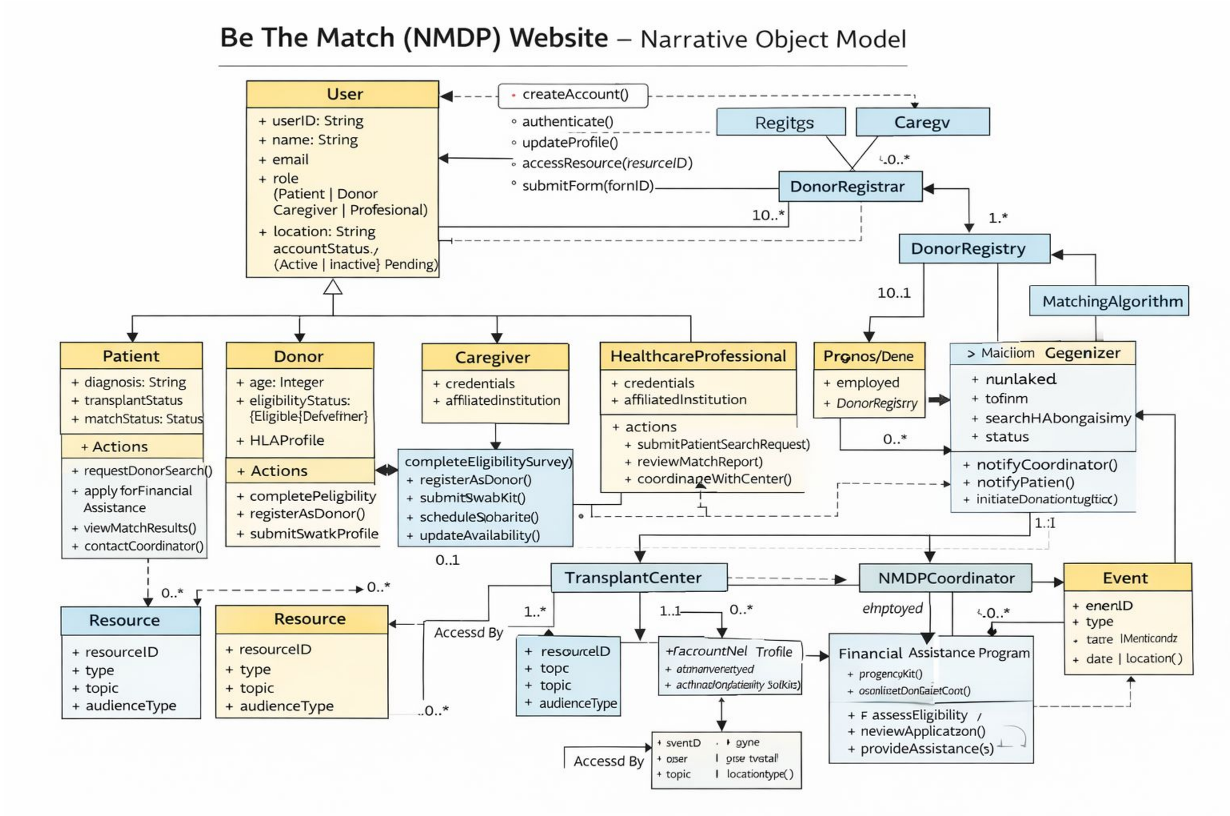


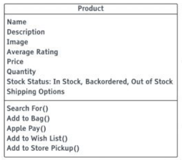
Figure 2.10

Narrative object model for an ATM. You can also view this diagram at objectmodelingfordesigners.com/chapter-2

ChatGPT Fail



Object Modeling Quick Reference



Objects are the nouns in a system.

Attributes characterize instances of an object.

Actions are the verbs.

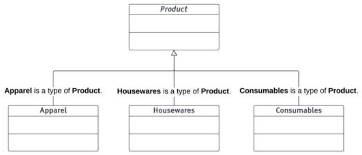
Relationships tell the structural story of the system, illustrating which objects are closely related and which have little or no relationship with one another.

Strictly speaking, all relationships are an association (indicated by a plain line). Aggregation, component, and inheritance are specific types of associations.

An **aggregation relationship** indicates an object that is a group or list of other objects. The line treatment for aggregation is an open diamond, with the diamond pointing at the aggregate object.

A **component relationship** is a type of dependency where one object is a component of another. The line treatment for a component is a filled-in diamond attached to the object that possesses the component or components.

Inheritance indicates a parent-child relationship between two objects where the child object inherits some or all of the parent's characteristics. The line treatment for inheritance relationships is an open triangle that points at the parent object.



Tips

Define What You Want To Achieve With the Model
Think about what you want to achieve with the model. Building a full model complete with all objects, attributes, actions, and relationships is not always necessary.

Go for Good Enough
Remember that the goal is not to create the perfect model; avoid the trap of analysis paralysis. It's OK not to have every corner of the model 100% worked out. Instead, focus on creating a model that serves your purpose—and not more.

Be Flexible in How You Build the Model
The first step in creating an object model is necessarily identifying the objects. You can build the other elements of the model (relationships, actions, attributes) in any order. Feel free to build the model in the way that makes the most sense to you or the given circumstances.

Have Fun!
Go into the modeling process with a sense of fun and adventure—to enjoy the discovery and clarity that object modeling can provide.

CLD to Stock & Flow

CLD 101

Causal Loop Diagrams (CLD) are great for:

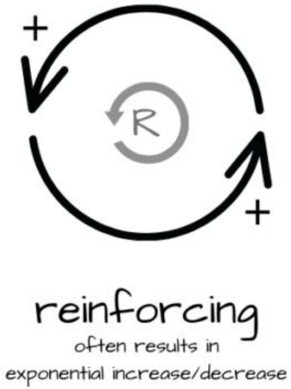
- “Quickly capturing your hypotheses about the causes of dynamics.
- Eliciting and capturing the mental models of individuals and teams.
- Communicating the important feedback processes you believe are responsible for a problem.

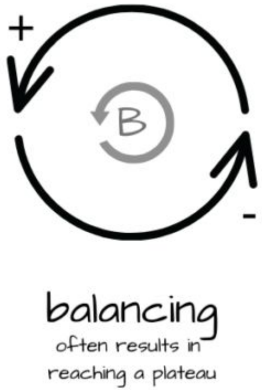
Figure 1.1 from [Müller, FMP, Springer 2015]

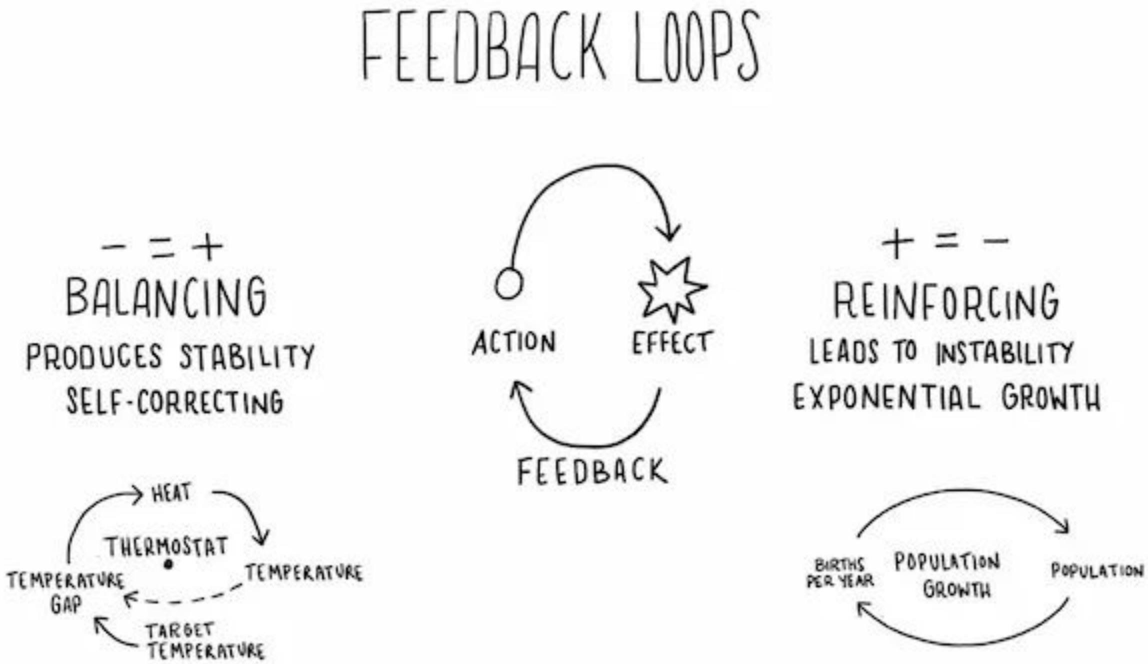
Feedback Loops

There are two kinds of feedback loops: reinforcing and balancing.

- In a **reinforcing loop**, the elements repeatedly perform the same action, resulting in exponential growth or decay.
- In a **balancing loop**, the elements of the system balance each other out.
- Both loops can bring along positive or negative consequences.





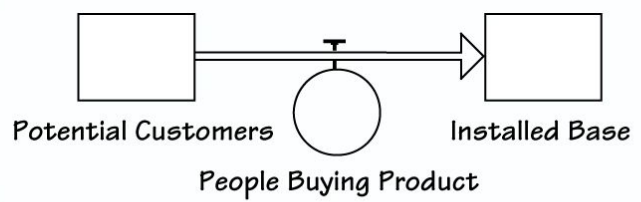

FEEDBACK LOOPS
− = +
BALANCING
PRODUCES STABILITY
SELF-CORRECTING
HEAT
THERMOSTAT
TEMPERATURE
TEMPERATURE GAP
TARGET TEMPERATURE
ACTION
EFFECT
FEEDBACK
+ = −
REINFORCING
LEADS TO INSTABILITY
EXPONENTIAL GROWTH
BIRTHS PER YEAR
POPULATION GROWTH
POPULATION

Cheat Sheet

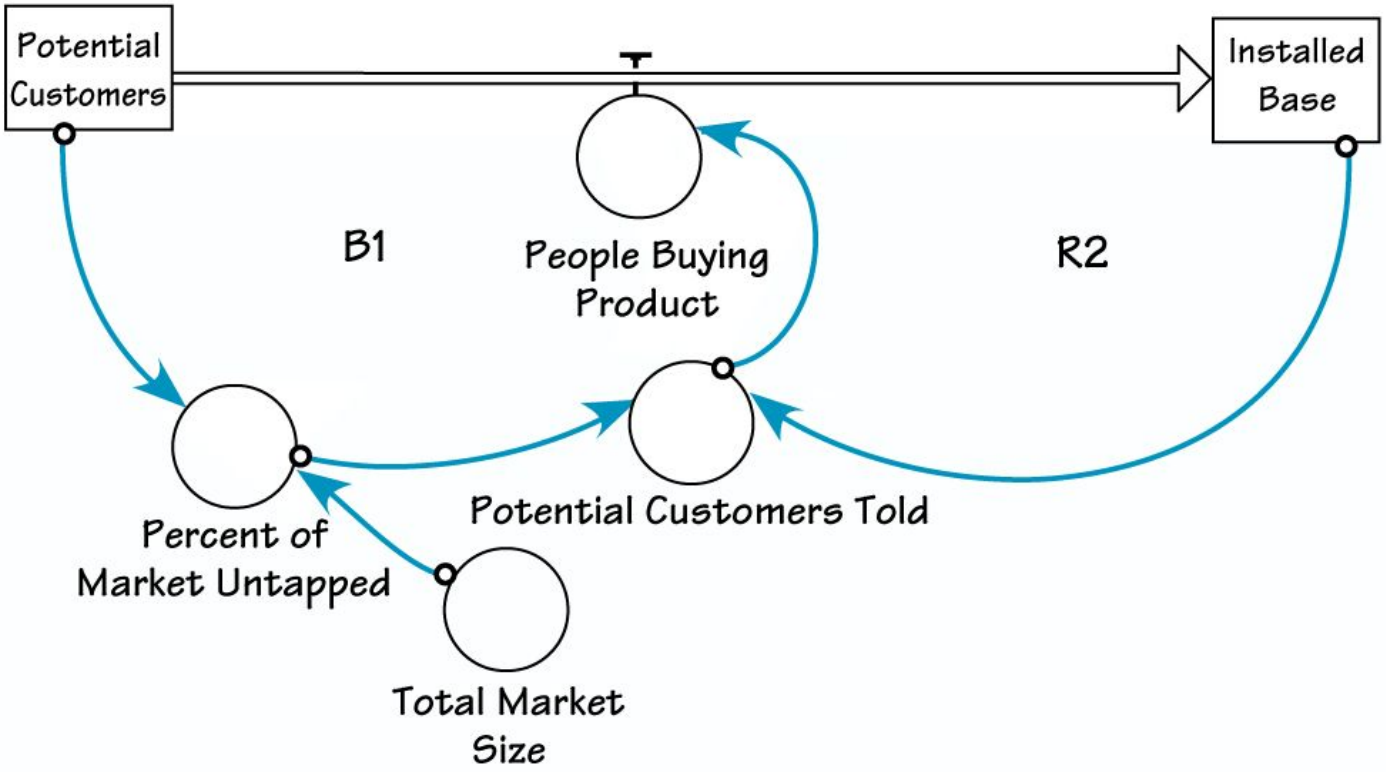
- A **causal loop diagram** consists of four basic elements: the variables, the links between them, the signs on the links (which show how the variables are interconnected), and the sign of the loop (which shows what type of behavior the system will produce).
- The first step in creating a causal “story” is to identify the nouns or variables that are important to the issue. Remember, a variable is something that can vary over time.
- Draw an arrow from the cause to the effect.
- Indicate an increase (draw a "+") or a decrease (draw a "–") for the second element in each relationship.
- Create the loop diagram by adding all the variables and linking them with s (same direction) or o (opposite direction) connections.
- To determine if a loop is reinforcing or balancing, one quick method is to count the number of “o’s.” If there are an even number of “o’s” (or none are present), the loop is reinforcing. If there are an odd number of “o’s,” it is a balancing loop.
- When designing a causal loop diagram, we focus on the accurate illustration of an individual’s beliefs instead of capturing the world as it is.
- Systems mapping helps us gain new insights by organizing experiences and motivations in a relational framework.

CLD to Stock & Flow

1) Stocks: Determine which CLD variables are stocks.
2) Flows: They are simply the variables that add to or subtract from the stocks. Only a flow can increase or decrease a stock, so if a variable is directly influencing a stock and is a function of time, it's a good bet that it's a flow.
3) Connect all flows to the stocks that they influence. If the flow has a negative effect on the stock, then it's an outflow; if it has a positive effect, it's an inflow.
4) Add any CLD variables that you did not previously identify as stocks or flows. These "auxiliary" variables are of two types: variables whose value does not change at all over the time period you are interested in — called "constants" — and variables that simply represent calculations based on stocks and flows.



First Iteration



Thank you!